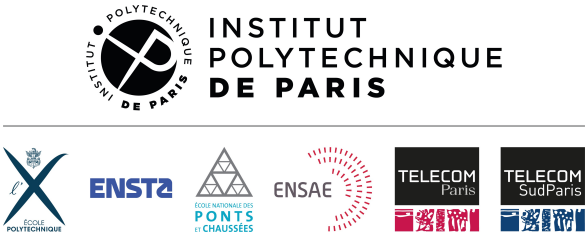

Deep Learning Course Project Report

MVA Master's

Transferability of DINOv3 Representations Under Extreme Domain Shift

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1 Introduction

The rise of self-supervised learning techniques that scale efficiently to models with billions of parameters has driven a paradigm shift in foundation models for computer vision. Among these, the DINO (self-**DI**stillation with **NO** labels) family of models [1, 2] has emerged as a powerful approach for learning visual representations without explicit supervision. The recently introduced DINOv3 [3] extends this framework with a 7 billion parameter model trained on 1.7 billion web images, incorporating a novel *Gram anchoring* objective designed to preserve patch-level feature consistency during extended training.

A key promise of such foundation models is their ability to provide high-quality features that transfer effectively to downstream tasks with minimal adaptation. The standard evaluation protocol—*linear probing*—involves training only a single linear layer on top of frozen backbone features. DINOv3 achieves impressive results on benchmarks like ADE20k semantic segmentation (55.9 mIoU) and ImageNet-1k classification (88.4% accuracy) using this approach.

However, these benchmarks predominantly consist of natural images similar to the training distribution. A critical question remains: **Do these improvements transfer to domains fundamentally different from natural web images?** This question has significant practical implications, as many real-world applications—medical imaging, remote sensing, marine biology—involve visual domains with substantially different characteristics.

In this work, we address this question by evaluating DINOv3 on three specialized domains representing *extreme distribution shifts*:

- **PatchCamelyon** [4]: Histopathology images of H&E-stained lymph node tissue for binary cancer classification (96×96 pixel patches).
- **DeepGlobe** [5]: Satellite imagery for 7-class land cover segmentation (2448×2448 pixel tiles).
- **Coralscapes** [6]: Underwater coral reef imagery for fine-grained 39-class benthic species segmentation (1024×2048 frames).

We first perform qualitative analysis of feature quality with cosine similarity maps and PCA visualizations, comparing DINOv2 to DINOv3, then conduct qualitative and quantitative analysis of DINOv3's performance on downstream tasks by comparing linear probing against Low-Rank Adaptation (LoRA) [7] fine-tuning.

2 Methods

2.1 Self-Supervised Learning with DINO

The DINO framework [1] learns visual representations through self-distillation between a student network f_θ and an exponential moving average (EMA) teacher network $f_{\theta'}$. Given an image x, multiple views (global and local crops) are generated, and the student is trained to match the teacher's output distribution on global crops:

$$\mathcal{L}_{\text{DINO}} = -\sum_{x \in \{x_1^g, x_2^g\}} \sum_{x' \in V \setminus x} p_{\theta'}(x) \log p_\theta(x') \tag{1}$$

where V denotes the set of all views and p_θ represents the softmax-normalized output.

DINOv2 [2] extends this with the iBOT [8] masked image modeling objective and KoLeo regularization [9] on class tokens, enabling joint optimization of global and local features.

2.2 Gram Anchoring in DINOv3

A key observation in scaling DINO models is that extended training improves performance on global tasks (e.g. classification) but degrades performance on dense prediction tasks (e.g. segmentation) [3]. This degradation manifests as loss of patch-level consistency: the cosine similarity maps between patch features become increasingly noisy over training iterations.

DINOv3 addresses this through *Gram anchoring*, a regularization that enforces consistency of the Gram matrix (pairwise patch similarities) without constraining individual feature values. Let $\mathbf{X}_S \in \mathbb{R}^{P \times d}$ denote the L_2-normalized patch features from the student network for an image with P patches, and $\mathbf{X}_G \in \mathbb{R}^{P \times d}$ the corresponding features from a *Gram teacher*—an earlier checkpoint with superior dense properties.

The Gram loss is:

$$\mathcal{L}_{\text{Gram}} = \left\| \mathbf{X}_S \cdot \mathbf{X}_S^\top - \mathbf{X}_G \cdot \mathbf{X}_G^\top \right\|_F^2 \tag{2}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. By operating on the Gram matrix $\mathbf{X} \cdot \mathbf{X}^\top$ rather than features directly, this loss preserves the *structure* of similarities while allowing features to evolve freely.

The full refinement objective combines all components:

$$\mathcal{L}_{\text{Ref}} = w_D \mathcal{L}_{\text{DINO}} + \mathcal{L}_{\text{iBOT}} + w_{DK} \mathcal{L}_{\text{DKoleo}} + w_{\text{Gram}} \mathcal{L}_{\text{Gram}} \tag{3}$$

2.3 Linear Probing Protocol and Low-Rank Adaptation (LoRA)

Following the DINOv3 evaluation protocol [3], we employ linear probing to assess feature quality. For image classification, we concatenate the CLS token with the mean-pooled patch tokens then train a single linear layer on top of this concatenated representation. For semantic segmentation, we employ a multi-scale feature fusion strategy. Given patch features $\{\mathbf{Z}^{(l)}\}_{l \in \mathcal{L}}$ extracted from multiple transformer layers $\mathcal{L}$, where each $\mathbf{Z}^{(l)} \in \mathbb{R}^{B \times C_l \times H' \times W'}$, we first concatenate them along the channel dimension then process them through dropout regularization, batch normalization, and a 1×1 convolution. We finally perform bilinear interpolation to restore the original spatial resolution (H, W), yielding class logits $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times K \times H \times W}$ for K semantic classes.

LoRA [7] provides parameter-efficient fine-tuning by injecting trainable low-rank matrices into frozen pretrained weights. For a pretrained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$, LoRA adds a low-rank decomposition: $\mathbf{W} = \mathbf{W}_0 + \Delta\mathbf{W} = \mathbf{W}_0 + \mathbf{BA}$ where $\mathbf{A} \in \mathbb{R}^{r \times k}$ and $\mathbf{B} \in \mathbb{R}^{d \times r}$ with rank $r \ll \min(d, k)$. We apply LoRA with rank $r = 16$, scaling factor $\alpha = 32$, and a dropout rate of 0.1 to the fused query-key-value projection (`qkv`), the output projection (`proj`), and both MLP fully connected layers (`fc1`, `fc2`) of the transformer.

3 Experimental Setup

3.1 Datasets

PatchCamelyon [4] comprises 327,680 histopathology patches (96×96 pixels) from H&E-stained lymph node sections. The binary classification task is to detect metastatic tissue. We use the provided train/validation/test split.

DeepGlobe [5] contains 1,146 satellite images (2448×2448 pixels) with 7 land cover classes: urban, agriculture, rangeland, forest, water, barren, and unknown. Since neither validation nor test labels are publicly available, we create our own 75/10/15 split from the 803 images of the training set.

Coralscapes [6] is the first general-purpose dense semantic segmentation dataset for coral reefs, comprising 2,075 underwater images (1,517 train/166 val/392 test, 1024×2048 pixels) from 35 dive sites across 5 countries in the Red Sea. Contains 174k annotated polygons over 39 benthic classes including various coral morphologies, algae-covered substrate, and other marine organisms. This represents the most challenging segmentation task due to high inter-class similarity and fine-grained taxonomic distinctions.

3.2 Implementation Details

We use the DINOv3 ViT-L/16 backbone (300M parameters) throughout our experiments. For linear probing, only the task-specific head is trained (2K–172K parameters depending on the task). For LoRA, we additionally train low-rank adaptations in all transformer blocks (∼6.3M trainable parameters in total). AdamW optimizer is used with a sequential learning rate schedule (linear warmup followed by cosine annealing). For classification, we use binary cross-entropy loss; for segmentation, a combination of cross-entropy and Jaccard loss. All experiments use early stopping with patience of 5–10 epochs. Our models are trained for only 16–30 epochs, compared to hundreds of epochs in reference methods.

4 Results

4.1 Qualitative Analysis: Feature Visualization

4.1.1 Cosine Similarity Maps

We visualize patch-level feature quality by computing cosine similarity between a reference patch (marked with a red cross) and all other patches. Ideally, semantically similar regions should have high similarity while distinct regions should have low similarity.

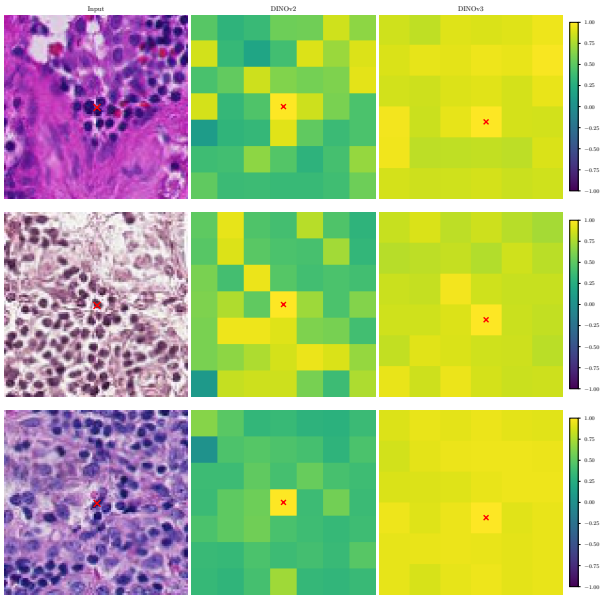

Figure 1: PatchCamelyon cosine similarity maps.

On PatchCamelyon (Figure 1), both DINOv2 (middle) and DINOv3 (right) struggle to produce spatially coherent similarity patterns on histopathology images. Notably, DINOv3 shows overly uniform responses (nearly all yellow), indicating loss of local discriminative power. The microscopic tissue patterns appear fundamentally different from natural image statistics. This suggests that the Gram anchoring objective, optimized for natural images, does not preserve local structure on this domain.

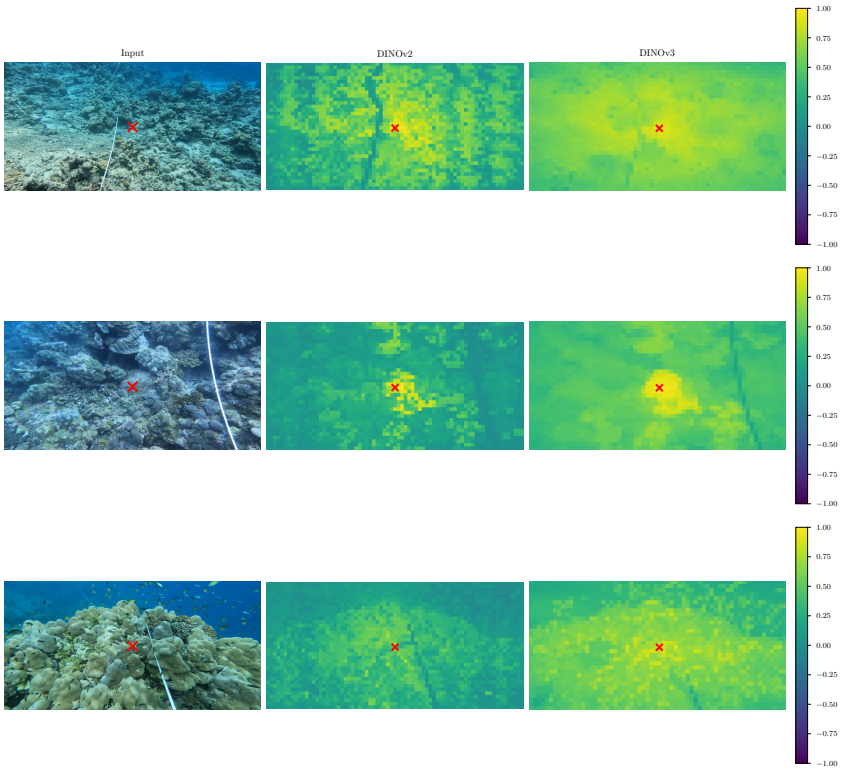

Figure 2: Coralscapes cosine similarity maps.

On Coralscapes (Figure 2), similarity maps show reasonable but modest localization on underwater imagery. DINOv2 demonstrates marginally better spatial coherence, suggesting the Gram anchoring improvements may not fully generalize to underwater imagery with its unique color distributions and lighting conditions.

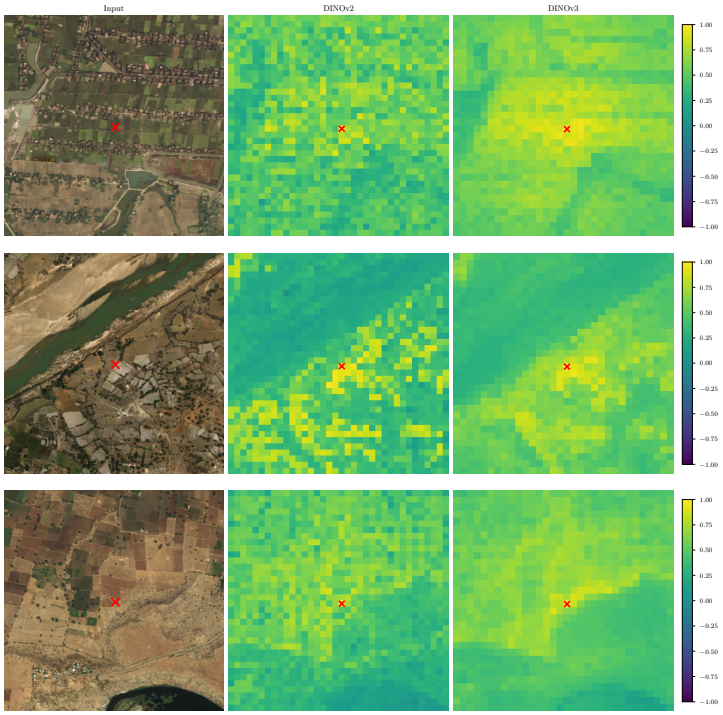

Figure 3: DeepGlobe cosine similarity maps.

On DeepGlobe (Figure 3), both models achieve reasonable localization, though DINOv2 maintains a slight advantage in local correspondence quality.

4.1.2 PCA Visualization

We project dense features to 3 dimensions via PCA and visualize as RGB images. This reveals global semantic structure independent of specific patch correspondences.

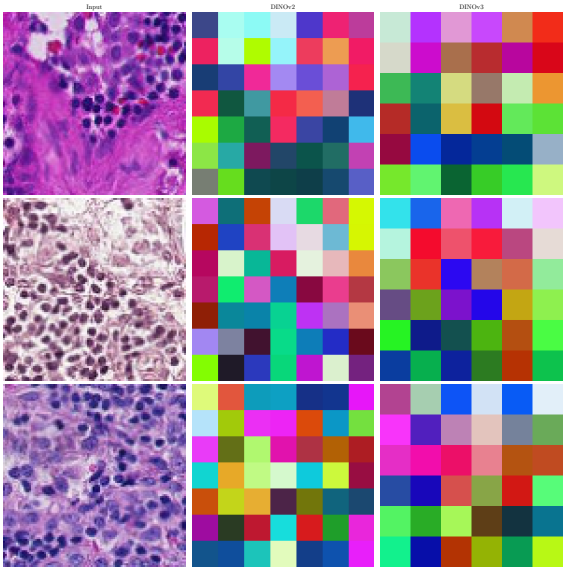

Figure 4: PatchCamelyon PCA visualization.

On PatchCamelyon (Figure 4), PCA does not reveal any meaningful semantic differentiation. Both models' outputs appear as random color patches, indicating that the learned features fail to capture the relevant structure in histopathology images. This confirms that histopathology represents the most challenging domain shift.

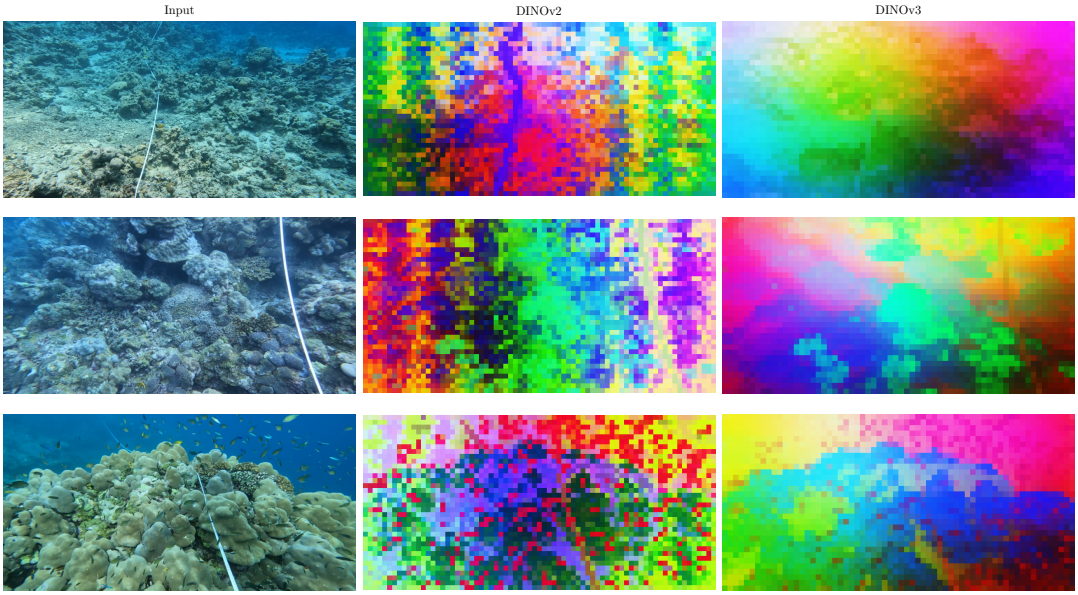

Figure 5: Coralscapes PCA visualization.

On Coralscapes (Figure 5), DINOv3 (right) demonstrates superior understanding of the underwater environment with smooth color gradients, better differentiation between coral species, substrate textures, and water. The enhanced feature quality from Gram anchoring appears beneficial for global semantic organization.

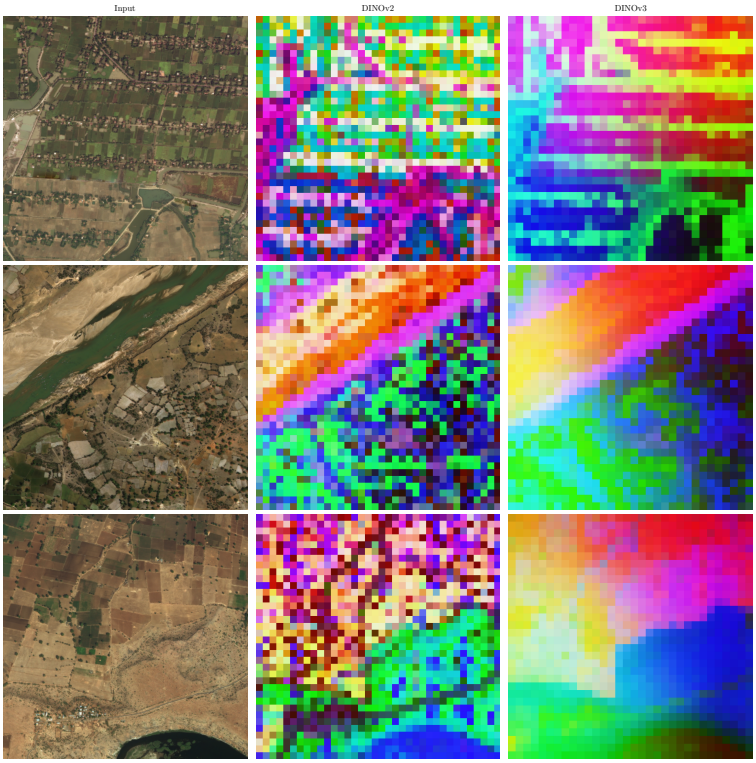

Figure 6: DeepGlobe PCA visualization.

On DeepGlobe (Figure 6), DINOv3 (right) produces dramatically smoother, semantically coherent feature maps. Color gradients align with land cover boundaries. DINOv2 (middle) features appear noisy and fragmented in comparison.

4.2 Qualitative Analysis: Segmentation Results

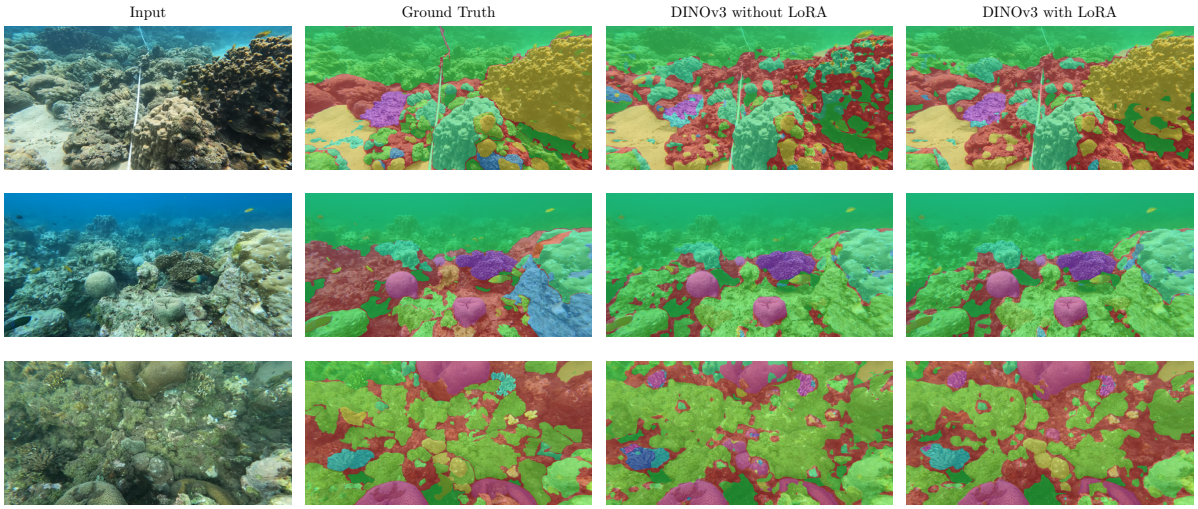

Figure 7: Coralscapes segmentation results.

Figure 7 shows that the 39-class fine-grained segmentation proves extremely challenging. Both configurations (DINOv3 with and without LoRA) struggle with species differentiation due to high visual similarity between benthic classes, though LoRA shows incremental gains.

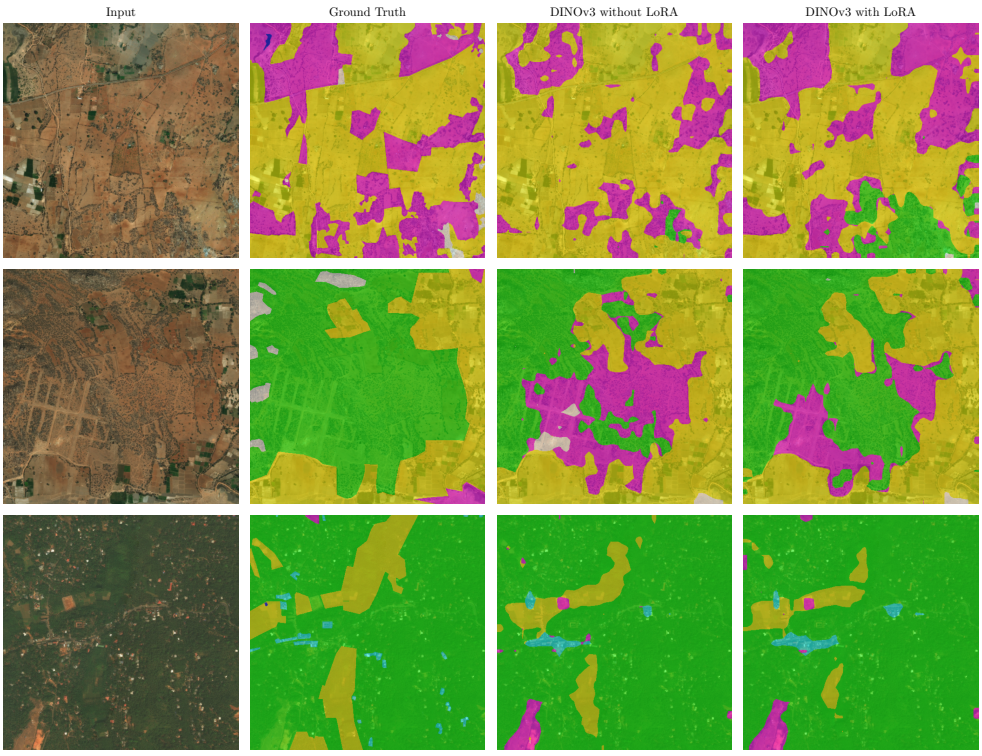

Figure 8: DeepGlobe segmentation results.

Figure 8 shows segmentation results on DeepGlobe. Both frozen backbone and LoRA-adapted DINOv3 models achieve reasonable land cover segmentation. LoRA provides modest improvements.

4.3 Quantitative Analysis

4.3.1 Classification: PatchCamelyon

Table 1 presents binary cancer classification results. Linear probing achieves 87% accuracy with only 2,049 trainable parameters. LoRA improves this to 92% accuracy and 0.973 AUC, exceeding the reference paper's reported 89.8% accuracy despite training for only 16 epochs versus their extensive optimization.

Table 1: PatchCamelyon binary classification results.

Method	Trainable Params	Accuracy	ROC AUC
DINOv3 Frozen + Linear	2,049	0.87	0.943
DINOv3 + LoRA + Linear	6,293,505	**0.92**	**0.973**
Reference [4]	119K	0.898	0.963

4.3.2 Semantic Segmentation

Table 2 summarizes segmentation performance. On DeepGlobe, linear probing achieves 0.66 mIoU, already exceeding the challenge's top score of 0.60 on their test set. LoRA further improves this to 0.69 mIoU.

Coralscapes proves more challenging: linear probing achieves 0.49 mIoU compared to the reference 0.578 mIoU. This suggests that the fine-grained 39-class problem with visually similar species requires a more sophisticated segmentation head.

Table 2: Semantic segmentation results. †Challenge test set differs from our split.

Dataset	Method	Params	mIoU	Pixel Acc
Coralscapes	DINOv3 Frozen + Linear	172,072	0.49	0.69
	DINOv3 + LoRA + Linear	6,463,528	0.51	0.71
	Reference [6]	—	0.578	0.828
DeepGlobe	DINOv3 Frozen + Linear	36,871	0.66	0.86
	DINOv3 + LoRA + Linear	6,328,327	**0.69**	**0.88**
	Challenge Top† [5]	—	0.60	—

5 Discussion

5.1 Key Findings

Linear probing with DINOv3 is remarkably efficient. With only 2K–172K trainable parameters, we achieve competitive results under extreme domain shift conditions, demonstrating that DINOv3 features encode substantial transferable knowledge.

Gram anchoring improves global semantic coherence. DINOv3's features show dramatically better organization in PCA visualizations, with smooth gradients aligning to semantic boundaries. However, this does not translate to improved local correspondence.

LoRA provides consistent gains. Across all tasks, LoRA fine-tuning yields 2–5% improvement with modest parameter overhead (∼6M), suggesting that some domain-specific adaptation is necessary for optimal performance.

DINOv3's transfer quality is domain-dependent. Satellite imagery (DeepGlobe) transfers best, exceeding the challenge's top models despite minimal training. Underwater imagery (Coralscapes) shows partial transfer. Histopathology (PatchCamelyon) represents the most challenging domain, with features appearing nearly random in visualizations.

5.2 Limitations

Our experiments use limited training epochs (16–30) compared to reference methods (often hundreds), and minimal hyperparameter tuning. While this demonstrates efficiency, further optimization could

improve results. Additionally, we only evaluate the ViT-L/16 variant; larger models might show different transfer characteristics.

6 Conclusion

Despite DINOv3's impressive benchmark results and novel Gram anchoring technique, linear probing alone is insufficient under extreme domain shift. The improvements from Gram anchoring, while significant on natural images, do not fully generalize to specialized domains with fundamentally different visual statistics. For such domains, either parameter-efficient fine-tuning (e.g., LoRA) or more sophisticated task-specific heads are necessary for optimal performance.

References

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